AGENTIC AI FOR ROBUST AND SOCIALLY AWARE AUTONOMOUS DRIVING

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Predicting pedestrian behavior is essential for safe and socially aware autonomous driving. This research proposes a Theory-of-Mind-inspired pedestrian behavioral-state prediction and risk-aware decision-making framework.

Using the Pedestrian Intention Estimation (PIE) dataset, 36 motion, pose, behavioral, and contextual features were extracted to classify pedestrians into six operational states: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk. LSTM and GRU models were combined through ensemble learning, achieving 71.5% accuracy, 62.1% macro-F1-score, and 72.4% weighted F1-score on the held-out test dataset.

Predicted behavioral states were integrated with Time-to-Collision (TTC)-based risk assessment to generate MAINTAIN, SLOW_DOWN, and STOP vehicle decisions. The framework was independently demonstrated in the CARLA simulation environment and presented through the Socialmind Drive web platform.

The results demonstrate the feasibility of combining pedestrian behavioral prediction with risk-aware decision-making, although further real-world validation is required.

Keywords: Autonomous Driving, Pedestrian Behavior, Theory of Mind, LSTM, GRU, Ensemble Learning, TTC

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
CARLA	Car Learning to Act
F1	F1-Score
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
PIE	Pedestrian Intention Estimation
RNN	Recurrent Neural Network
SLIIT	Sri Lanka Institute of Information Technology
ToM	Theory of Mind
TTC	Time-to-Collision

1 INTRODUCTION

1.1 Overall Project Context

Autonomous driving systems must operate in dynamic environments where safe navigation depends on understanding the behavior and intentions of surrounding road users. Urban traffic presents challenges because pedestrians, cyclists, and drivers may act unpredictably, communicate intentions indirectly, and respond differently to traffic conditions [1], [2].

The overall research project, “Agentic AI for Robust and Socially Aware Autonomous Driving,” investigates the use of agentic artificial intelligence to improve autonomous vehicle decision-making in complex urban environments. The project consists of four independently developed research components addressing different challenges.

The first component, Adaptive Intent Prediction in Occluded Urban Scenarios, investigates road-user intention prediction under partial visibility. The second component, Multi-Agent Negotiation for Right-of-Way in Complex Intersections, explores interaction and negotiation among autonomous vehicles. The third component, Proactive Social Compliance Modeling Using Theory of Mind, focuses on estimating pedestrian behavioral states to support socially aware driving decisions. The fourth component, Federated Collective Learning Between Autonomous Vehicles Using Agentic Knowledge Sharing, investigates collaborative learning and knowledge sharing among autonomous vehicles.

This individual thesis focuses on the third component, Proactive Mental State Prediction Modeling using Theory of Mind. It investigates pedestrian behavioral-state prediction using observable motion, body posture, behavioral patterns, and contextual information. The proposed framework combines temporal deep learning models with risk-aware decision-making to support interpretable autonomous vehicle responses.

The component is developed and evaluated independently, contributing to the broader research objective of socially aware autonomous driving.

1.2 Individual Component Context

Pedestrian behavior prediction is essential for autonomous vehicles operating in urban environments. Conventional approaches commonly focus on pedestrian crossing intention, movement patterns, and trajectory estimation. However, pedestrians may display behavioral cues such as hesitation, distraction, sudden movement, or changes in posture that provide additional information about possible actions [1], [2].

Theory of Mind (ToM) refers to the ability to attribute mental states, including intentions, beliefs, and desires, to other individuals [8]. Inspired by this concept, the proposed research investigates whether observable pedestrian characteristics can support the estimation of six operational behavioral states: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk.

The research uses the Pedestrian Intention Estimation (PIE) dataset and MediaPipe-based pose estimation to extract 36 motion, pose, behavioral, and contextual features [1], [6]. Temporal relationships are analyzed using Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) neural networks. Their predictions are combined through an ensemble approach to estimate pedestrian behavioral states.

The predicted states are integrated with Time-to-Collision (TTC)-based risk assessment and predefined decision rules to generate vehicle responses, including MAINTAIN, SLOW_DOWN, and STOP. The integrated framework is demonstrated in the CARLA autonomous driving simulator [4].

The six categories are treated as operational behavioral approximations rather than directly verified psychological states. The primary focus is to investigate how these predictions can contribute to context-aware and risk-sensitive autonomous vehicle decision-making.

1.3 Problem Statement

Autonomous vehicles must anticipate pedestrian behavior to make appropriate decisions in complex traffic environments. Existing pedestrian prediction approaches primarily investigate crossing intentions, movement patterns, and future trajectories. Although these methods provide useful information, they may not adequately

distinguish behavioral characteristics such as hesitation, distraction, sudden crossing commitment, or unpredictable movement [1], [2].

Pedestrians with similar observable movements may exhibit different behavioral tendencies. For example, a pedestrian standing near a crossing may be waiting, hesitant, or preparing to enter the roadway. Distinguishing these situations is challenging because internal intentions cannot be directly observed and must be estimated from movement, posture, and environmental context.

Furthermore, pedestrian prediction results are not always explicitly connected to interpretable vehicle decision-making. Identifying a pedestrian's crossing intention alone may provide insufficient information for selecting an appropriate response under uncertain traffic conditions [7].

Therefore, the central research problem is the limited integration of detailed pedestrian behavioral-state estimation with proactive, socially aware, and risk-sensitive autonomous vehicle decision-making.

This research investigates whether a Theory-of-Mind-inspired framework can estimate multiple pedestrian behavioral states from observable evidence and incorporate these estimates into interpretable vehicle decisions.

1.4 Research Gap

Existing autonomous driving research has made significant progress in pedestrian detection, crossing intention prediction, and trajectory estimation. Studies such as PIE and PSI demonstrate the importance of understanding pedestrian movement, crossing behavior, and interactions with surrounding traffic [1], [2]. However, predicting observable actions alone may not provide sufficient information about the underlying behavioral states that influence pedestrian decisions.

Recent studies have explored pose-based motion estimation, context-aware prediction, and spatial–temporal learning techniques to improve pedestrian intention prediction [3], [5], [9]. Although these approaches contribute to understanding pedestrian movements and crossing intentions, there remains a need to investigate how subtle behavioral cues can be used to distinguish more detailed states such as hesitation, commitment, distraction, aggression, and jaywalking. Pedestrians

exhibiting similar movement patterns may have different behavioral intentions, creating uncertainty for autonomous vehicle decision-making.

Theory of Mind (ToM) provides a conceptual basis for reasoning about the intentions and mental states of other agents [8]. However, translating this concept into autonomous driving remains challenging because pedestrian mental states cannot be directly observed. Further research is needed to determine how observable body movements, behavioral patterns, and environmental context can support reliable approximations of these states.

Furthermore, existing approaches do not always explicitly connect detailed pedestrian behavioral-state estimates with socially aware vehicle decision-making. Understanding whether a pedestrian is hesitant, distracted, or committed to crossing could provide additional context for anticipating potential conflicts and selecting appropriate vehicle responses [2], [7].

Therefore, the research gap addressed by this study is the limited integration of Theory-of-Mind-inspired, multi-class pedestrian behavioral-state inference with proactive, socially aware autonomous vehicle decision-making. This research investigates the gap between conventional pedestrian intention prediction and more detailed behavioral understanding, with the aim of supporting context-sensitive, interpretable, and risk-aware driving decisions.

1.5 Aim and Objectives

The aim of this research is to develop and evaluate a Theory-of-Mind-inspired pedestrian behavioral-state prediction framework that infers latent behavioral states from observable pedestrian cues and supports proactive, socially aware, and risk-sensitive decision-making in autonomous vehicles.

Research Objectives:

- To investigate existing pedestrian intention prediction and Theory-of-Mind-inspired approaches to identify their limitations in understanding pedestrian behavioral states and supporting socially aware autonomous driving.

- To identify and extract relevant pedestrian motion, body posture, behavioral, and environmental contextual features that can provide evidence of underlying pedestrian behavioral states.
- To develop a temporal prediction framework capable of classifying pedestrians into six behavioral categories: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk.
- To evaluate the effectiveness of pedestrian behavioral-state prediction using appropriate classification metrics and examine the contribution of combining temporal prediction models.
- To integrate the predicted pedestrian behavioral states into a risk-aware autonomous vehicle decision-making mechanism that considers potential pedestrian–vehicle conflicts.
- To evaluate the integrated framework in a simulated urban driving environment and examine its ability to support interpretable and socially aware vehicle responses.

1.6 Research Questions or Hypotheses

The primary research question of this study is: How can a Theory-of-Mind-inspired pedestrian behavioral-state prediction framework support proactive, socially aware, and risk-sensitive decision-making in autonomous vehicles?

To address this question, the research investigates whether observable pedestrian motion, body posture, behavioral patterns, and environmental context can provide sufficient information to distinguish between behavioral states such as waiting, hesitation, commitment, distraction, aggression, and jaywalking.

The study also examines how effectively temporal deep learning models can classify these pedestrian behavioral states and whether combining predictions from LSTM and GRU models improves classification performance compared with individual models.

Another question concerns how predicted pedestrian behavioral states can be incorporated into risk-aware vehicle decision-making. In particular, the study investigates how behavioral-state predictions, together with Time-to-Collision (TTC)

information, can inform vehicle responses such as maintaining speed, slowing down, or stopping.

Finally, the research examines how the proposed framework performs in simulated urban driving scenarios and what limitations arise when translating pedestrian behavioral-state predictions into autonomous vehicle decisions.

1.7 Scope Limitations and Assumptions

Scope

This research focuses on Proactive Social Compliance Modeling Using Theory of Mind, specifically the prediction of pedestrian behavioral states and their application to autonomous vehicle decision-making. The study investigates six behavioral categories: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk. It considers pedestrian motion, body posture, behavioral patterns, and environmental context as observable evidence for estimating these states.

The scope includes pedestrian behavioral analysis using the Pedestrian Intention Estimation (PIE) dataset, temporal prediction using LSTM and GRU models, and the evaluation of their combined predictions. It also covers the integration of predicted behavioral states with risk-aware decision logic and the examination of vehicle responses in the CARLA simulation environment. The study is limited to the individual Theory-of-Mind-inspired component and does not implement the occlusion prediction, multi-agent negotiation, or federated knowledge-sharing components of the overall research project.

Limitations

A major limitation is that pedestrian mental states cannot be directly observed or verified from motion and posture alone. Therefore, the six predicted categories represent behavioral approximations rather than confirmed psychological states. In addition, the available pedestrian dataset was not originally designed to provide ground-truth annotations for all six categories, which introduces uncertainty into the assigned labels and their interpretation.

The prediction performance may also be affected by class imbalance, limited representation of unusual pedestrian behaviors, and variations in pedestrian

appearance, camera perspective, or environmental conditions. Furthermore, the use of simulation restricts the extent to which the findings can be generalized to real-world autonomous driving. The research does not establish improved road safety through real-vehicle testing or demonstrate adaptation to different regional traffic cultures.

Assumptions

The research assumes that observable pedestrian movements, body posture, and surrounding contextual information contain useful cues for estimating behavioral states. It also assumes that temporal changes in these cues can provide additional information compared with isolated observations.

The study further assumes that the available pedestrian data and derived behavioral labels provide a suitable experimental basis for investigating the proposed classification problem, while acknowledging that these labels do not represent directly verified mental states. For simulation-based evaluation, it is assumed that the configured pedestrian and vehicle scenarios provide a controlled environment for examining the relationship between predicted behavioral states and autonomous vehicle responses.

Within these boundaries, the research aims to evaluate the feasibility and limitations of Theory-of-Mind-inspired pedestrian behavioral-state estimation as a supporting component of socially aware autonomous driving.

1.8 Individual Contribution

The individual contribution of this research is the development and evaluation of a Theory-of-Mind-inspired pedestrian behavioral-state prediction and risk-aware decision-making framework.

The research formulates a six-class operational behavioral-state classification problem consisting of Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk. These categories provide a more detailed behavioral representation than conventional binary crossing intention prediction, while remaining approximations of observable behavior.

A structured feature-processing pipeline was developed using 36 motion, pose, behavioral, and contextual features. Temporal patterns were analyzed using independently trained LSTM and GRU models, and their predictions were combined through an equal-weight probability-averaging ensemble.

The research also developed a TTC-based decision-making mechanism that connects predicted pedestrian states with interpretable MAINTAIN, SLOW_DOWN, and STOP vehicle responses.

Finally, the prediction and decision-making components were integrated into a standalone CARLA simulation prototype and examined through selected single-pedestrian and multi-pedestrian scenarios.

The main contribution is the investigation of a computational connection between multi-class pedestrian behavioral-state estimation and risk-aware autonomous vehicle decision-making, rather than direct measurement of psychological states or verified improvements in real-world driving safety.

1.9 Thesis Structure

This thesis is organized into nine chapters, covering the research background, methodology, system development, evaluation, and conclusions.

Chapter 1 - Introduction introduces the research background, problem statement, objectives, research questions, scope, and contributions.

Chapter 2 - Literature Review and Related Work reviews previous studies on pedestrian behavior prediction, Theory of Mind, pose estimation, deep learning, and autonomous driving.

Chapter 3 - Research Methodology explains the research methods, dataset preparation, feature extraction, model training, and evaluation techniques.

Chapter 4 - Product and Solution Design describes the proposed system architecture, main components, data flow, and decision-making framework.

Chapter 5 - Implementation, Integration and Testing explain the implementation of prediction models, ensemble approach, decision-making system, and CARLA simulation testing.

Chapter 6 - Results and Validation presents the performance of the LSTM, GRU, and ensemble models, along with CARLA simulation results.

Chapter 7 - Discussion discusses the research findings, challenges, limitations, and their relevance to socially aware autonomous driving.

Chapter 8 - Conclusion and Future Directions summarizes the research findings, contributions, achievement of objectives, and recommendations for future improvements.

The thesis ends with references and appendices containing supporting sources and additional technical information.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

This chapter reviews existing research related to pedestrian intention prediction, behavioral-state estimation, Theory of Mind (ToM), and socially aware autonomous driving. The review aims to identify the contributions and limitations of previous approaches and establish the theoretical and technical foundations for the proposed research.

A thematic review approach was adopted, focusing on five areas: pedestrian crossing intention and trajectory prediction, pose-based behavioral analysis, temporal deep learning, Theory-of-Mind-inspired reasoning, and risk-aware autonomous vehicle decision-making. Relevant studies were examined according to their prediction objectives, input features, computational methods, and applicability to pedestrian–vehicle interactions.

The review primarily focuses on the individual research component, Proactive Mental State Prediction Modeling using Theory of Mind. The other components of the overall research project are outside its detailed scope.

2.2 Domain and Theoretical Foundations

Autonomous vehicles must anticipate the behavior of surrounding road users to make appropriate decisions in dynamic traffic environments. Pedestrians are particularly challenging because their movements may change unexpectedly, and their intentions cannot always be determined from current position or trajectory alone. Existing research has therefore investigated crossing intention prediction, trajectory estimation, and contextual behavioral analysis [1], [2].

Theory of Mind

Theory of Mind refers to the ability to attribute mental states, such as beliefs, desires, and intentions, to other individuals [8]. In autonomous driving, this concept provides a theoretical basis for interpreting pedestrian behavior beyond directly observable movement.

For example, pedestrians standing near a crossing may be waiting, hesitant, or preparing to cross despite appearing stationary. Computational Theory-of-Mind approaches, including inverse planning, investigate how another agent's possible goals can be inferred from observed actions [10].

The proposed research adopts a Theory-of-Mind-inspired perspective by estimating six operational pedestrian behavioral states: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk. These categories represent observable behavioral approximations rather than independently verified psychological states.

Pedestrian Intention and Behavioral Prediction

The Pedestrian Intention Estimation (PIE) dataset introduced by Rasouli et al. supports research on pedestrian crossing intention and trajectory prediction using visual and contextual observations [1]. Similarly, the PSI dataset investigates pedestrian behavior and social interactions relevant to autonomous driving [2].

Although these resources support the study of pedestrian intentions, conventional crossing prediction does not necessarily distinguish behavioral characteristics such as hesitation, distraction, or sudden commitment. More detailed behavioral representations may therefore provide additional information for vehicle decision-making.

Pose and Temporal Behavioral Analysis

Pedestrian body posture and movement patterns provide useful evidence for behavioral interpretation. Research on pedestrian motion-state estimation from two-dimensional pose demonstrates the relevance of body configuration and movement characteristics [5]. Context-aware pedestrian prediction further emphasizes the importance of environmental information [3].

Temporal analysis is also necessary because pedestrian behavior changes over time. Sequential models can examine transitions between standing, hesitation, and crossing, while spatial–temporal methods such as PedAST-GCN demonstrate the value of learning movement relationships across observations [9].

These findings support the use of motion, pose, behavioral, and contextual features combined with temporal prediction models.

Risk-Aware Autonomous Vehicle Decision-Making

Behavioral prediction becomes useful for autonomous driving when its outputs contribute to appropriate vehicle responses. Interaction-aware decision-making approaches consider surrounding road users and potential conflicts when determining driving actions [7].

Time-to-Collision (TTC) is a commonly used indicator for assessing the urgency of potential traffic conflicts [19], [20]. Combining TTC with estimated pedestrian behavior provides a basis for selecting interpretable responses such as maintaining speed, slowing down, or stopping.

The proposed framework therefore connects pedestrian behavioral-state prediction with TTC-based risk assessment and rule-based vehicle decision-making.

2.3 Related Research

Previous research has investigated pedestrian behavior prediction using visual observations, contextual information, temporal learning, and interaction-aware reasoning.

Pedestrian Crossing Intention Prediction

Rasouli et al. introduced PIE as a dataset and research foundation for pedestrian crossing intention and trajectory prediction [1]. EfficientPIE, proposed by Qu et al., investigates computationally efficient pedestrian crossing intention prediction using limited observations [11].

Graph-based and attention-based methods have also been developed to model pedestrian movement. GTransPDM employs graph-embedded transformer techniques for crossing intention prediction [12], while PedAST-GCN uses spatial–temporal attention and graph convolutional networks [9].

These approaches demonstrate the importance of temporal and contextual information. However, their primary prediction targets differ from the six operational behavioral categories investigated in this study.

Pose-Based Pedestrian Behavior Analysis

Research on pedestrian motion-state estimation using two-dimensional pose investigates how body configuration can support movement interpretation [5]. MonoPIC proposes a monocular pedestrian intention classification framework intended for low-latency applications [13].

Context-aware prediction approaches also investigate the combination of pedestrian movement and environmental information [3]. These studies motivate the use of multiple observable cues rather than relying exclusively on trajectory information.

Nevertheless, pose-based movement recognition and crossing intention classification should not be interpreted as direct measurements of psychological mental states.

Theory-of-Mind-Inspired Behavioral Reasoning

Baker et al. investigated action understanding through inverse planning, providing a computational perspective on inferring goals from observed behavior [10].

Within autonomous driving, StylePredict applies a machine Theory-of-Mind perspective to human driver behavior prediction using trajectories [14]. Zhang et al. examined pedestrian crossing intentions during driver–pedestrian negotiations through temporal segmentation [15].

These studies demonstrate the relevance of reasoning about the possible intentions of other road users. However, they address different prediction tasks and do not directly validate the six-class pedestrian behavioral-state representation used in this research.

Explainable and Socially Aware Autonomous Driving

Munir et al. investigated pedestrian intention prediction using a vision-language model capable of producing interpretable information [16]. Varga et al. explored interaction-aware model predictive decision-making for socially compliant autonomous driving in mixed urban traffic [7].

These approaches highlight the importance of connecting pedestrian understanding with interpretable driving decisions. The present study investigates this connection through a different architecture that combines temporal behavioral classification with predefined TTC-based decision rules.

Synthesis of Related Research

The reviewed studies demonstrate progress in crossing intention prediction, pose-based analysis, temporal learning, and interaction-aware autonomous driving. However, they employ different datasets, prediction targets, and evaluation procedures.

The proposed research investigates whether a six-class behavioral representation derived from observable pedestrian features can support interpretable, risk-aware vehicle decisions. Its contribution concerns the integration of behavioral-state estimation and decision support rather than direct recognition of human mental states.

2.4 Existing Systems Products and Techniques

Existing approaches relevant to the proposed research can be grouped into five technical categories.

Pedestrian intention prediction systems: PIE, EfficientPIE, GTransPDM, and PedAST-GCN investigate crossing intention or trajectory prediction using visual, contextual, graph-based, and temporal information [1], [9], [11], [12]. These methods provide useful foundations but do not directly address the same six-class classification problem.

Pose estimation and behavioral analysis: MediaPipe supports the extraction of body landmarks from visual observations [6]. Pose-based pedestrian motion analysis and MonoPIC demonstrate how body configuration and movement features can contribute to pedestrian prediction [5], [13].

Temporal deep learning: Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks process sequential observations and capture temporal dependencies. These characteristics make them suitable for examining changes in pedestrian movement and behavioral patterns. Attention-based and graph-based methods provide alternative approaches to temporal prediction [9], [12].

Theory-of-Mind-inspired techniques: Computational approaches such as inverse planning, StylePredict, and pedestrian–driver negotiation modeling investigate how

observed actions can provide evidence about possible goals or intentions [10], [14], [15].

Risk-aware decision-making systems: Interaction-aware vehicle decision-making approaches incorporate road-user behavior and potential conflicts into driving decisions [7]. TTC-based assessment provides an interpretable measure of conflict urgency [19], [20], while CARLA offers a controlled simulation environment for examining pedestrian–vehicle interactions [4].

These techniques inform the proposed framework, which combines structured pedestrian features, LSTM–GRU temporal prediction, and TTC-based decision-making within a standalone simulation-based prototype.

2.5 Gap Synthesis and Conceptual Framework

The reviewed literature demonstrates significant progress in pedestrian crossing intention prediction, trajectory estimation, pose-based analysis, and socially aware autonomous vehicle decision-making. However, several research gaps remain when these approaches are considered in relation to proactive pedestrian behavioral understanding.

Gap Synthesis

The first identified gap concerns the limited scope of pedestrian behavior classification. Existing approaches, including PIE, EfficientPIE, GTransPDM, and PedAST-GCN, primarily investigate pedestrian crossing intentions or future movement [1], [9], [11], [12]. Although these predictions are valuable for autonomous driving, a binary crossing decision does not fully represent behavioral differences such as waiting, hesitation, distraction, or commitment. This motivates the investigation of a more detailed behavioral-state representation.

The second gap concerns the integration of multiple observable behavioral cues. Pose-based techniques demonstrate the usefulness of body configuration, while context-aware approaches emphasize the influence of environmental conditions [3], [5], [13]. However, the reviewed studies differ in how they combine motion, posture, temporal behavior, and contextual information. This creates an opportunity to

investigate whether a unified feature representation can support the classification of multiple pedestrian behavioral states.

The third gap relates to temporal behavioral understanding. Pedestrian behavior can change rapidly as individuals approach or interact with a roadway. Spatial–temporal prediction methods capture movement patterns over time [9], [12], but their evaluation objectives do not directly establish performance on the six behavioral categories considered in this research. Consequently, further investigation is required to determine how effectively sequential learning models can distinguish these categories.

The fourth gap concerns the connection between Theory-of-Mind-inspired behavioral inference and autonomous vehicle decision-making. Existing research has investigated machine Theory of Mind, pedestrian–driver negotiation, and socially compliant driving [7], [14], [15]. Nevertheless, these studies address different agents, prediction targets, and decision-making objectives. The reviewed literature therefore motivates examining a framework that connects multi-class pedestrian behavioral estimates with interpretable, risk-sensitive vehicle responses.

Conceptual Framework

To address these gaps, the proposed conceptual framework consists of three connected stages: pedestrian behavioral observation, temporal behavioral-state estimation, and risk-aware vehicle decision-making.

The first stage involves collecting observable pedestrian information from visual and contextual data. Motion characteristics, body posture, behavioral patterns, and environmental context provide the input features used to represent pedestrian activity. These features are treated as measurable indicators rather than direct observations of psychological states.

The second stage involves analyzing temporal sequences of pedestrian features using Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models. Their predictions are combined through an ensemble approach to estimate six operational behavioral categories: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk. This stage represents the Theory-of-Mind-inspired component of the

framework because it attempts to infer behavioral tendencies from observable evidence.

The third stage connects predicted behavioral states with risk assessment. Time-to-Collision (TTC) and rule-based decision logic are used to determine an appropriate vehicle response, such as maintaining speed, slowing down, or stopping. The framework is examined within the CARLA simulation environment to investigate its operation in controlled pedestrian–vehicle interaction scenarios [4].

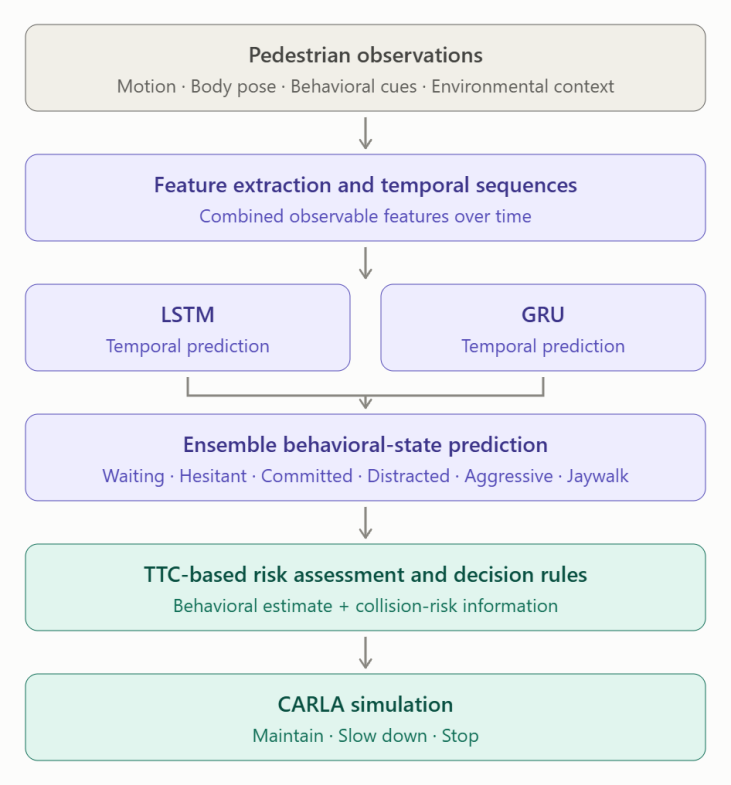


Figure 1: Conceptual framework

Research Positioning

The conceptual framework does not assume that pedestrian mental states can be directly identified from visual information. Instead, it investigates whether observable behavioral indicators can support a useful approximation of pedestrian behavioral states. Similarly, simulation-based evaluation is intended to examine model performance and decision behavior rather than establish real-world safety improvements.

Overall, the identified research gap is the need to investigate an integrated approach that combines multi-cue pedestrian observation, temporal multi-class behavioral-state estimation, and interpretable risk-aware vehicle decision-making. The proposed framework addresses this gap by connecting Theory-of-Mind-inspired behavioral reasoning with autonomous driving decisions while explicitly recognizing the uncertainty and limitations of inferred pedestrian states.

3 RESEARCH METHODOLOGY

3.1 Research Design

This research adopts a quantitative, experimental, and simulation-based research design to investigate the feasibility of Theory-of-Mind-inspired pedestrian behavioral-state estimation for socially aware autonomous driving. The study focuses on developing and evaluating a computational framework that uses observable pedestrian behavior to estimate behavioral states and support risk-aware vehicle decision-making.

The research follows a model development and evaluation approach consisting of data preparation, feature extraction, temporal model training, ensemble prediction, and simulation-based integration. The Pedestrian Intention Estimation (PIE) dataset is used as the primary data source because it provides pedestrian observations and contextual information relevant to crossing behavior [1]. Since the original dataset does not directly provide the six behavioral-state labels required by this study, operational labels are derived from observable behavioral indicators using predefined rules. These labels are treated as behavioral proxies rather than verified psychological mental states.

The proposed framework represents pedestrian behavior using four groups of observable features: motion, body posture, behavioral patterns, and environmental context. Body pose information is extracted using MediaPipe, while movement and contextual characteristics are processed into structured numerical features [6]. The resulting observations are organized into temporal sequences to capture changes in pedestrian behavior over time.

Two recurrent neural network architectures, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), are independently trained to classify six pedestrian behavioral categories: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk. Their prediction probabilities are subsequently combined through an equal-weight ensemble to investigate whether combining the models improves classification performance. The models are evaluated using a held-out test set and classification metrics, including accuracy, precision, recall, and F1-score. Attention

is given to class-level performance because overall accuracy alone may conceal differences in prediction reliability across behavioral categories.

The research also incorporates a simulation-based evaluation stage using CARLA [4]. Predicted pedestrian behavioral states are integrated with Time-to-Collision (TTC) assessment and predefined decision rules to produce vehicle responses such as maintaining speed, slowing down, or stopping. Controlled simulation scenarios are used to examine how behavioral predictions influence the decision-making process under different pedestrian-vehicle interaction conditions.

The overall research design is experimental because it involves implementing, testing, and comparing computational models under defined evaluation conditions. It is quantitative because performance is assessed through numerical classification measures and simulation observations. The simulation component provides a controlled environment for examining system behavior without introducing risks to real road users.

The study is limited to evaluating the proposed framework using available pedestrian data and simulated driving scenarios. It does not directly measure human psychological states or establish the safety of the system in real-world traffic. Within these boundaries, the research design provides a structured basis for assessing the performance, feasibility, and limitations of integrating temporal pedestrian behavioral-state prediction with risk-aware autonomous vehicle decision-making.

3.2 Data Participants or Study Materials

This research uses secondary pedestrian observation data, machine learning models, and simulation tools to investigate Theory-of-Mind-inspired pedestrian behavioral-state estimation. As the study is computational and simulation-based, no human participants are directly recruited, interviewed, or involved in experimental trials. Instead, the research relies on an existing pedestrian dataset and controlled virtual driving scenarios.

Primary Dataset

The primary data source is the Pedestrian Intention Estimation (PIE) dataset introduced by Rasouli et al. [1]. The dataset contains real-world urban traffic video

recordings and pedestrian annotations, including information relevant to pedestrian movement, crossing behavior, and surrounding traffic conditions. These observations provide the foundation for extracting features associated with pedestrian behavior over time.

Although PIE supports pedestrian intention prediction, it does not directly contain the six behavioral-state labels required for this research. Therefore, the categories Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk are derived using predefined behavioral rules and observable indicators. These categories represent operational behavioral proxies rather than independently verified psychological mental states. Consequently, the resulting labels are suitable for evaluating the proposed classification framework but cannot establish that the model accurately identifies a pedestrian's internal thoughts or intentions.

Pedestrian Features and Data Preparation

The study uses four groups of pedestrian features: motion, body posture, behavioral patterns, and environmental context. Motion features describe changes in pedestrian position and movement, while posture features represent body configuration extracted from visual observations. Behavioral features capture patterns such as movement consistency and hesitation-related indicators, whereas contextual features describe relevant information about the surrounding environment.

MediaPipe is used to obtain pedestrian pose landmarks from image frames [6]. These observations are processed into a structured representation containing 36 features: 10 motion features, 12 pose features, 9 behavioral features, and 5 contextual features. The extracted features are organized into 30-frame temporal sequences to preserve temporal information relevant to pedestrian behavior.

The prepared data is divided into training, validation, and testing subsets using a 70:15:15 allocation. Pedestrian tracks are separated across the subsets to reduce the risk of data leakage between training and evaluation. Feature scaling is fitted using training data and subsequently applied to the validation and testing subsets to maintain consistent preprocessing.

Machine Learning Study Materials

The principal machine learning materials consist of Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) neural network models. Both architectures are used to analyze sequential pedestrian features and classify the six operational behavioral categories. An equal-weight ensemble combines their predicted class probabilities to produce the final behavioral-state estimate.

The study also uses Python-based data processing and machine learning libraries for feature preparation, model training, evaluation, and inference. Model performance is assessed using classification accuracy, precision, recall, F1-score, and confusion matrices.

Simulation Environment

CARLA is used as the simulation environment for examining pedestrian–vehicle interactions under controlled conditions [4]. The simulation incorporates pedestrian scenarios and vehicle responses to investigate how predicted behavioral states can inform driving decisions.

The decision-making component combines behavioral-state predictions with Time-to-Collision (TTC) assessment and predefined rules. Depending on the estimated behavioral state and collision-risk conditions, the vehicle may maintain its current movement, slow down, or stop. These scenarios provide a controlled basis for observing the operation of the integrated framework.

Data Considerations and Limitations

The use of an existing dataset avoids direct human participant recruitment. However, the study remains dependent on the quality, representativeness, and annotation characteristics of the original recordings. Pose estimation errors, incomplete observations, and class imbalance may affect the extracted features and subsequent predictions.

A further limitation is that the six behavioral categories are generated from observable indicators rather than direct psychological assessments. The evaluation therefore measures agreement with the operational labeling scheme, not the validity of inferred human mental states. Similarly, CARLA-based observations provide

evidence about system operation in simulation but do not independently demonstrate real-world driving safety.

Overall, the selected dataset, extracted features, temporal learning models, and simulation tools provide the study materials required to investigate multi-class pedestrian behavioral-state estimation and its integration with risk-aware autonomous vehicle decision-making.

3.3 Experimental or Study Procedure

The experimental procedure followed a structured sequence of dataset preparation, feature extraction, behavioral-state labeling, model development, performance evaluation, and simulation-based testing. Each stage was designed to investigate the feasibility of predicting pedestrian behavioral states from observable information and integrating these predictions into risk-aware autonomous vehicle decision-making.

Dataset Preparation and Preprocessing

The experiment began with the Pedestrian Intention Estimation (PIE) dataset, which contains pedestrian observations recorded in real-world traffic environments [1]. Pedestrian trajectories, associated annotations, and video frames were processed to obtain information relevant to pedestrian movement and crossing behavior. The data was organized according to pedestrian tracks to preserve the temporal relationship between consecutive observations.

Pose Estimation and Feature Extraction

Pedestrian body landmarks were extracted from video frames using MediaPipe [6]. The extracted pose information was combined with motion, behavioral, and contextual observations to construct a 36-dimensional feature representation. These features were used to capture observable pedestrian movement, posture, hesitation patterns, and environmental conditions relevant to crossing behavior [1], [9].

Table 1: Summary of Extracted Pedestrian Features

Feature Category	Description	Number of Features
Motion	Speed, acceleration, deceleration, step characteristics, and movement pauses	10
Pose	Body angles, head orientation, foot angles, body leaning, and orientation changes	12
Behavioral	Hesitation, pause duration, curb proximity, looking direction, and direction changes	9
Contextual	Traffic lights, vehicle distance, crosswalk presence, road width, and pedestrian density	5
Total	Numerical features per frame	36

The extracted features provided structured numerical inputs for the LSTM and GRU models. The behavioral indicators represented observable patterns rather than direct measurements of pedestrians' internal mental states.

Behavioral-State Label Generation

Since the PIE dataset does not directly provide the six behavioral-state labels required for this study, a rule-based labeling procedure was applied to derive operational categories from observable pedestrian information. The six categories were Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk.

The labeling procedure used predefined behavioral indicators and available contextual information to assign categories to pedestrian observations. These labels were treated as behavioral proxies rather than direct measurements of psychological mental states. This distinction was maintained throughout model development and evaluation.

Temporal Sequence Construction and Dataset Splitting

The extracted features were organized into temporal windows of 30 frames, with 36 numerical features per frame, resulting in an input shape of 30 × 36. A sliding-

window stride of five frames was used to generate temporal samples. This procedure enabled the models to examine changes in pedestrian behavior across consecutive observations rather than relying on individual frames.

The prepared dataset was divided into training, validation, and testing subsets using a track-disjoint 70:15:15 allocation. Pedestrian tracks were kept separate across these subsets to reduce data leakage. Feature normalization was performed using a scaler fitted on the training data, and the same transformation was applied to validation and testing observations.

Model Development and Training

Two recurrent neural network architectures, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), were developed independently for six-class pedestrian behavioral-state classification. Both models received the same temporal feature representation, allowing their predictive performance to be examined under a common data preparation procedure.

The models were trained using the training subset, while the validation subset supported model selection and training assessment. Following individual model development, an equal-weight ensemble was constructed by averaging the class probability outputs of the LSTM and GRU models. The behavioral category with the highest combined probability was selected as the ensemble prediction.

Performance Evaluation

The trained LSTM, GRU, and ensemble models were evaluated using the held-out testing subset. Classification accuracy, precision, recall, F1-score, and confusion matrices were used to examine predictive performance.

Performance was assessed both overall and across individual behavioral categories. Attention was given to differences between classes because class imbalance and overlapping behavioral indicators could affect classification reliability. The ensemble was compared with the individual models to determine whether combining their predictions improved the evaluation results.

Integration with Risk-Aware Decision-Making

Following model development, the behavioral-state prediction component was connected to a rule-based vehicle decision mechanism. The mechanism combined predicted pedestrian behavioral states with Time-to-Collision (TTC) estimates to assess potential pedestrian–vehicle conflicts.

Based on the estimated behavioral state and TTC conditions, the decision mechanism generated responses such as maintaining speed, slowing down, or stopping. This integration was intended to investigate how behavioral-state predictions could contribute to interpretable vehicle decisions rather than relying exclusively on pedestrian distance or movement information.

Simulation-Based Testing

The integrated framework was examined using the CARLA autonomous driving simulator [4]. Controlled pedestrian–vehicle interaction scenarios were used to observe behavioral predictions, TTC assessments, and corresponding vehicle responses.

Simulation outputs were examined to determine whether the prediction and decision-making components operated together as intended. The evaluation focused on system behavior within the simulated environment and did not assume that the observed responses established real-world safety improvements.

Overall, the experimental procedure provided a structured approach for examining pedestrian behavioral-state classification and its integration with autonomous vehicle decision-making. The findings were interpreted in relation to the limitations of derived behavioral labels, prediction uncertainty, and simulation-based evaluation.

3.4 Measures and Evaluation Metrics

The proposed framework was evaluated using quantitative classification metrics and simulation-based observations. The evaluation focused on three main aspects: pedestrian behavioral-state classification performance, comparative performance of the Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and ensemble models, and the operation of risk-aware vehicle decision-making within the CARLA simulation environment.

Classification Performance Metrics

The predictive performance of the LSTM, GRU, and ensemble models was evaluated using accuracy, precision, recall, and F1-score. These metrics are widely used to assess the performance of classification systems and provide complementary information about prediction reliability [17].

Accuracy measures the proportion of correctly classified samples among all evaluated observations. Precision represents the proportion of predictions assigned to a particular behavioral category that are correct, while recall measures the proportion of actual samples belonging to that category that are correctly identified. The F1 score combines precision and recall through their harmonic meaning, providing a balanced measure of classification performance [17].

Both macro-average and weighted-average F1-scores were considered. Macro-F1 calculates the average F1-score across all behavioral categories while assigning equal importance to each class. Weighted F1 considers the number of samples belonging to each category. These measures are particularly relevant because imbalanced class distributions may cause overall accuracy to conceal poor performance on less frequently represented behavioral categories [18].

Confusion Matrix Analysis

Confusion matrices were used to examine the distribution of correct and incorrect predictions across the six behavioral categories: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk. This analysis supported the identification of misclassification patterns between categories with similar observable characteristics, particularly those involving stationary movement, hesitation, or crossing-related behavior.

Class-specific precision, recall, and F1-scores were also examined to identify differences in prediction reliability. These measures provided a more detailed assessment of model performance than overall accuracy alone [17].

Comparative Model Evaluation

The LSTM and GRU models were independently evaluated using the same held-out testing dataset. Their performance was compared using accuracy, macro-F1, weighted F1, and class-level classification results.

An equal-weight ensemble was subsequently evaluated by averaging the class probability predictions generated by the LSTM and GRU models. The ensemble results were compared with those of the individual models to investigate whether combining their predictions improved classification performance.

The evaluation considered improvements in accuracy and F1-score separately because higher overall accuracy does not necessarily indicate better performance across all behavioral categories. Attention was given to macro-F1 to assess the balance of classification performance across the six classes.

Risk Assessment and Decision-Making Measures

Time-to-Collision (TTC) was used as a risk assessment measure to estimate the remaining time before a potential pedestrian–vehicle conflict under the assumptions of the calculation. TTC is an established traffic safety indicator used to assess the severity of potential collisions [19].

Van der Horst and Hogema [20] reported that traffic encounters with a minimum TTC below 1.5 seconds are generally considered critical. This finding provided a research basis for selecting 1.5 seconds as the critical-risk threshold within the proposed framework.

The implemented decision-making mechanism used a TTC threshold of 2.5 seconds for warning-level conditions and 1.5 seconds for critical conditions. The 2.5-second warning threshold was adopted as an implementation parameter to enable an earlier response before the critical condition was reached. These thresholds were combined with predicted pedestrian behavioral states to generate vehicle responses, including maintaining speed, slowing down, and stopping.

The selected thresholds were applied to the simulation-based decision-making framework and were not assumed to represent universally validated braking limits. In particular, the 1.5-second value was supported as a critical traffic-conflict indicator

rather than an independently validated optimal emergency-braking activation threshold [20].

Simulation-Based Evaluation

The integrated framework was examined using the CARLA autonomous driving simulator [4]. Controlled pedestrian–vehicle interaction scenarios were used to observe the relationship between predicted behavioral states, TTC estimates, and corresponding vehicle responses.

The simulation-based evaluation considered behavioral-state predictions, associated confidence values, TTC measurements, and the actions generated by the decision-making mechanism. These observations were used to examine whether the prediction and decision components operated together as intended and whether the resulting actions were consistent with the implemented decision rules.

The simulation assessment focused on the operational behavior of the integrated framework. It did not establish statistically validated reductions in collision frequency, improvements in passenger comfort, or real-world autonomous driving safety.

Evaluation Limitations

The six pedestrian behavioral categories were derived from observable indicators rather than independently verified psychological assessments. Consequently, classification accuracy, precision, recall, and F1-score measured agreement with the operational labeling scheme rather than the ability to identify actual internal mental states.

Furthermore, the classification and simulation evaluations represented different aspects of system performance. Improvements in behavioral-state classification did not automatically demonstrate safer vehicle decisions, while appropriate responses within simulated scenarios did not establish real-world effectiveness.

Overall, the selected evaluation measures provided a structured basis for assessing multi-class pedestrian behavioral-state prediction, comparing the LSTM and GRU

models with their ensemble, and examining the integration of behavioral predictions with TTC-based risk-aware autonomous vehicle decision-making

3.5 Data Analysis

The data analysis process followed a quantitative approach to evaluate the performance of the proposed Theory-of-Mind-inspired Pedestrian Behavioral-state prediction framework. The analysis focused on examining the extracted pedestrian features, evaluating the LSTM and GRU classification models, assessing the effectiveness of their ensemble, and interpreting the behavior of the integrated risk-aware decision-making mechanism.

Data Preparation and Feature Analysis

The extracted pedestrian observations were represented using 36 numerical features grouped into four categories: motion, body posture, behavioral patterns, and environmental context. These features were organized into temporal sequences of 30 frames to preserve changes in pedestrian behavior over time.

Before model evaluation, the prepared dataset was examined for class distribution and feature consistency. Track-disjoint training, validation, and testing split was used to reduce data leakage, while feature scaling parameters were obtained from the training subset and applied consistently to the validation and testing subsets.

The distribution of the six operational behavioral categories: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk was considered when interpreting the classification results. This was important because differences in class frequency could influence model performance and the reliability of overall accuracy as an evaluation measure [18].

Classification Performance Analysis

The LSTM and GRU models were analyzed independently using the held-out testing dataset. Classification accuracy, precision, recall, and F1-score were calculated to assess effectively each model distinguished from the six behavioral categories. These measures provided complementary perspectives on classification performance [17].

Macro-averaged and weighted-average F1-scores were examined alongside overall accuracy. Macro-F1 was used to assess performance across all classes with equal importance, while weighted F1 reflected differences in class frequency. Confusion matrices and class-specific results were analyzed to identify frequently misclassified behavioral categories and potential similarities between their observable features.

Ensemble Performance Analysis

The predictions generated by the LSTM and GRU models were combined through an equal-weight probability-averaging ensemble. The ensemble was evaluated using the same testing dataset and performance metrics as the individual models.

A comparative analysis was conducted to determine whether the ensemble improved overall accuracy, macro-F1, or weighted F1 relative to the individual models. Differences between these metrics were interpreted separately, since an improvement in overall accuracy does not necessarily indicate more balanced classification across all behavioral categories.

The analysis also considered the limitations of combining models that may share similar prediction errors. Therefore, ensemble performance was interpreted using the observed evaluation results rather than assuming that model combination would consistently improve classification reliability.

Risk Assessment and Decision Analysis

The decision-making component was analyzed using predicted pedestrian behavioral states and Time-to-Collision (TTC) estimates. TTC was used to represent the urgency of potential pedestrian–vehicle conflicts, with lower values indicating more immediate risk under the assumptions of the calculation [19].

The implemented decision rules used a warning-level TTC threshold of 2.5 seconds and a critical threshold of 1.5 seconds. The critical threshold was informed by research identifying minimum TTC values below 1.5 seconds as indicative of critical traffic encounters [20]. The warning threshold was treated as an implementation parameter for initiating an earlier response.

The resulting vehicle actions, including maintaining speed, slowing down, and stopping, were examined to determine whether they were consistent with the predicted behavioral states, TTC conditions, and predefined decision rules.

Simulation Data Analysis

The integrated framework was examined through controlled pedestrian–vehicle interaction scenarios in CARLA [4]. Simulation outputs, including predicted behavioral states, prediction confidence, TTC estimates, and generated vehicle responses, were reviewed to assess the operation of the complete prediction and decision-making pipeline.

The analysis focused on whether the framework produced interpretable responses under the simulated conditions and whether the decision mechanism behaved consistently with its implemented logic. The simulation observations were not used to claim statistically established collision reduction or improvements in real-world driving safety.

Interpretation of Findings

The classification and simulation findings were interpreted in relation to the research objectives and the limitations of the experimental design. Particular attention was given to class imbalance, similarities between behavioral categories, and the reliability of predictions derived from observable pedestrian information.

Because the behavioral labels were generated using predefined rules rather than independent psychological assessments, the classification results were interpreted as measures of agreement with the operational labeling scheme. They were not considered direct evidence that the models could accurately identify pedestrians' internal mental states.

Overall, the data analysis process provided a structured basis for comparing temporal classification models, assessing ensemble performance, and examining how predicted pedestrian behavioral states contributed to risk-aware autonomous vehicle decision-making within the simulation environment.

3.6 Ethics Privacy Security and Responsible AI

Ethical considerations, privacy protection, system security, and responsible artificial intelligence are important aspects of autonomous driving research, particularly when machine learning models are used to interpret human behavior and support vehicle decisions. This study considered these aspects throughout data preparation, model development, evaluation, and simulation-based testing.

Ethical Considerations

The research used the existing Pedestrian Intention Estimation (PIE) dataset rather than directly recruiting human participants or conducting experiments involving road users [1]. The experimental activities were conducted using previously collected pedestrian observations and simulated pedestrian–vehicle interactions. Therefore, the study did not involve direct intervention in real traffic or exposing participants to risks arising from experimental vehicle behavior.

A central ethical consideration was the interpretation of pedestrian behavioral states. The proposed framework classified pedestrians into six categories: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk. These categories were derived from observable movement, posture, behavioral patterns, and contextual information rather than direct assessments of psychological conditions.

Consequently, the classifications were treated as operational behavioral proxies and not as verified descriptions of a pedestrian's thoughts, personality, or intentions. Categories such as Aggressive and Distracted were used only within the defined computational labeling scheme and were not intended to make personal or moral judgments about individuals.

Privacy and Data Protection

The research relied on an existing pedestrian dataset containing observations from public traffic environments. Although no new personal information was collected directly from participants, the use of video-based pedestrian data still raises privacy considerations because recorded images may contain identifiable individuals.

To reduce unnecessary processing of personal information, the prediction framework primarily used structured numerical features derived from pedestrian movement,

body landmarks, behavioral indicators, and environmental context. The study did not require facial recognition, personal identity classification, or the identification of individual pedestrians.

Dataset materials and extracted information were intended for research purposes, subject to the applicable dataset access and usage conditions. Any future public release of processed data or demonstration recordings should be reviewed to avoid exposing unnecessary identifiable information.

System Security and Reliability

Security and reliability are important because incorrect or manipulated pedestrian observations could influence behavioral predictions and subsequent vehicle decisions. Potential technical risks include missing pose landmarks, inaccurate feature extraction, corrupted input sequences, and incorrect behavioral-state predictions.

The framework addressed aspects of data reliability through consistent preprocessing, feature normalization, and separation of training and testing observations. However, these measures do not establish protection against adversarial attacks, unauthorized access, or deliberate manipulation of sensor information.

The study was conducted in a controlled research and simulation environment. It did not claim to provide a production-ready cybersecurity architecture for autonomous vehicles. Additional security testing would be required before considering deployment in safety-critical driving systems.

Responsible AI and Model Fairness

Responsible AI considerations were particularly relevant because the framework attempted to infer pedestrian behavioral states from observable information. Such predictions are inherently uncertain, and different pedestrians may exhibit similar movements for different reasons.

The operational labels were generated using predefined rules rather than independently verified psychological assessments. As a result, the models could

learn patterns associated with the labeling procedure without necessarily capturing a pedestrian's true intention. This limitation was acknowledged when interpreting classification results.

Potential sources of bias included class imbalance, variation in pedestrian movement, differences in visibility and pose-estimation quality, and the representativeness of the PIE dataset. Classification performance was therefore examined using class-specific precision, recall, F1-scores, and confusion matrices rather than relying solely on overall accuracy [17], [18].

The study did not establish that the framework performs equally well across different demographic groups, geographical regions, or cultural settings. Such conclusions would require additional representative datasets and dedicated fairness evaluations.

Transparency and Explainability

The framework was designed to provide an interpretable connection between predicted behavioral states, TTC-based risk assessment, and vehicle decisions. The rule-based decision mechanism allowed vehicle responses, such as maintaining speed, slowing down, or stopping, to be examined in relation to predefined conditions.

However, the use of interpretable decision rules does not make the internal reasoning of the LSTM and GRU models fully explainable. Prediction confidence was also not treated as proof that a behavioral classification was correct.

Safety and Responsible Deployment

The integrated framework was evaluated using the CARLA simulator, which enabled pedestrian–vehicle interaction scenarios to be examined without introducing experimental risks to real road users [4]. TTC-based decision rules were used to investigate potential vehicle responses under controlled conditions.

The critical TTC threshold of 1.5 seconds was informed by traffic-conflict research [20], while the 2.5-second warning threshold was adopted as an implementation parameter. Neither threshold was treated as a universally applicable guarantee of safe vehicle operation.

The research did not involve deployment on public roads or demonstrate validated real-world collision reduction. Before practical deployment, further evaluation would be necessary to examine robustness, uncertainty handling, cybersecurity, fairness, and compliance with applicable vehicle safety requirements.

Overall, the study adopted a responsible research perspective by limiting experiments to existing data and simulation, avoiding claims of direct psychological-state recognition, and acknowledging the limitations of behavioral labels and predictive models. These considerations establish appropriate boundaries for interpreting the findings and identify ethical, privacy, security, and safety requirements for future development.

3.7 Reproducibility and Research Artefacts

Reproducibility is an essential aspect of experimental machine learning research, as it enables researchers to examine the procedures, verify reported findings, and extend existing work. In this study, reproducibility was supported through the maintenance of source code, data processing procedures, trained models, evaluation scripts, and simulation-related artefacts. These materials document the development and evaluation of the proposed Theory-of-Mind-inspired pedestrian behavioral-state prediction framework.

Data Processing and Feature Extraction Artefacts

The Pedestrian Intention Estimation (PIE) dataset was used as the primary source of pedestrian observations [1]. The data processing workflow included feature extraction, behavioral-state labeling, sequence generation, and feature normalization.

The feature extraction process produced 36 numerical features consisting of 10 motion features, 12 pose features, nine behavioral features, and five contextual features. MediaPipe-based pose estimation was incorporated to extract pedestrian body landmarks [6]. The extracted observations were organized into temporal sequences of 30 frames using a sliding-window stride of five frames.

The dataset was divided into training, validation, and testing subsets using a track-disjoint 70:15:15 split to reduce information leakage between subsets. A feature scaler fitted on the training data was preserved and reused during evaluation and

inference. Maintaining the same preprocessing procedure is important for ensuring consistency between model training and subsequent predictions.

Model Training and Prediction Artefacts

The research maintained separate implementation scripts and experiment notebooks for the Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models. These artefacts document the development of the temporal classification models used to predict six operational pedestrian behavioral categories: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk.

The trained models were saved in Keras format as `lstm_mental_state_best.keras` and `gru_mental_state_best.keras`. The fitted feature scaler was stored as `scaler.pkl`. These files support reuse of the trained models without requiring the complete training process to be repeated.

The ensemble implementation combined the class probability outputs of the LSTM and GRU models using equal-weight averaging. The corresponding evaluation script supports comparison of the ensemble against the individual models under the same testing conditions.

Evaluation and Experimental Artefacts

The experimental evaluation was supported by scripts and notebooks used to assess model performance through accuracy, precision, recall, macro-F1, weighted F1, and confusion matrices. These metrics provide a structured basis for evaluating multi-class classification performance [17].

The use of macro-averaged and weighted-average metrics was particularly relevant because differences in behavioral-category frequencies could affect the interpretation of overall accuracy [18]. Retaining the evaluation procedures enables future researchers to examine classification performance and identify potential differences between individual models and the ensemble.

Simulation and Decision-Making Artefacts

The integration of behavioral prediction with autonomous vehicle decision-making was implemented using the CARLA simulation environment [4]. The research

artefacts included scripts for pedestrian pose processing, model inference, ensemble prediction, rule-based decision-making, and simulation scenario execution.

The decision-making mechanism used predicted pedestrian behavioral states together with Time-to-Collision (TTC) estimates to determine vehicle actions, including maintaining speed, slowing down, and stopping. The implementation applied a 2.5-second warning threshold and a 1.5-second critical threshold. The latter was informed by established traffic-conflict research [20].

The simulation scripts and execution logs provide supporting material for examining the interaction between behavioral predictions, risk assessment, and vehicle decisions under controlled experimental conditions.

Research Artefact Management and Reproducibility Limitations

The principal research artefacts consisted of Python source files, Jupyter notebooks, saved neural network models, a serialized preprocessing scaler, simulation scenarios, and execution logs. Together, these materials support the examination and potential reproduction of the implemented framework.

Nevertheless, exact reproduction of experimental results may depend on factors such as dataset availability, software library versions, hardware configurations, random initialization, and differences in the simulation environment. Complete independent reproduction would therefore require the relevant dependencies, execution configurations, and dataset preparation instructions to be documented alongside the research artefacts.

Additionally, the six behavioral categories were derived from operational labeling rules rather than independently verified psychological assessments. Reproducing the experimental results would require consistent application of these labeling rules, and the resulting performance measures should not be interpreted as direct validation of pedestrians' internal mental states.

Overall, the maintained research artefacts provide a foundation for transparency, verification, and future extension of the proposed framework. They document the main stages of pedestrian feature extraction, temporal behavioral-state classification, ensemble prediction, and TTC-based vehicle decision-making while acknowledging

the limitations associated with reproducibility in machine learning and simulation-based research.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The proposed solution was designed to support proactive and socially aware autonomous driving by predicting pedestrian behavioral states and incorporating these predictions into risk-aware vehicle decision-making. The system requirements were identified based on the research objectives, characteristics of the Pedestrian Intention Estimation (PIE) dataset, and the need to integrate pedestrian behavior prediction with the CARLA simulation environment [1], [4].

The requirements were categorized into functional requirements, non-functional requirements, and technical requirements to guide the design and implementation of the proposed framework.

Functional Requirement

Functional requirements describe the main operations that the proposed system must perform.

- Pedestrian Data Processing: The system must process pedestrian observations obtained from the PIE dataset and prepare structured information for behavioral-state prediction. This includes extracting motion, pose, behavioral, and contextual information [1].
- Pose and Feature Extraction: The system must extract 36 numerical features comprising 10 motion features, 12 pose features, nine behavioral features, and five contextual features. MediaPipe-based pose estimation must support the extraction of relevant body landmarks [6].
- Temporal Sequence Generation: The system must organize extracted features into 30-frame temporal sequences to capture changes in pedestrian behavior over time. Each sequence must contain 36 features per frame, producing an input dimension of 30 × 36.
- Pedestrian Behavioral-State Prediction: The system must classify pedestrian observations into six operational behavioral categories: Waiting, Hesitant,

Committed, Distracted, Aggressive, and Jaywalk, using trained LSTM and GRU models.

- Ensemble Prediction: The system must combine the class probability outputs of the LSTM and GRU models through equal-weight averaging to generate a final behavioral-state prediction.
- Risk Assessment: The system must estimate pedestrian–vehicle interaction risk using Time-to-Collision (TTC). The implemented decision mechanism must distinguish warning-level conditions at 2.5 seconds and critical conditions at 1.5 seconds. The critical threshold is informed by established traffic-conflict research [19], [20].
- Vehicle Decision-Making: The system must use predicted pedestrian behavioral states and TTC estimates to generate one of three vehicle responses: MAINTAIN, SLOW_DOWN, or STOP.
- Simulation Integration: The system must integrate the behavioral prediction and decision-making components with CARLA to examine pedestrian–vehicle interactions under controlled simulation scenarios [4].

Non-Functional Requirements

Non-functional requirements describe the quality characteristics expected from the system.

- Prediction Performance: The system should provide reliable behavioral-state classification across all six categories. Performance must be evaluated using accuracy, precision, recall, macro-F1, weighted F1, and confusion matrices [17].
- Computational Efficiency: The prediction pipeline should support repeated processing of pedestrian observations during simulation. Computational efficiency is important for responsive decision-making, although strict real-time performance guarantees are outside the validated scope of this study.
- Reliability and Consistency: The system should apply consistent feature ordering, normalization, and temporal sequence formatting during training, testing, and inference. This is necessary to reduce inconsistencies between model development and simulation integration.

- Interpretability: The system should provide observable behavioral predictions, confidence values, TTC estimates, and corresponding vehicle actions. These outputs should make the operation of the decision-making mechanism understandable during evaluation.
- Modularity: The system should maintain separate components for feature extraction, behavioral prediction, ensemble processing, risk assessment, and vehicle control. This modular structure supports testing, maintenance, and future improvements.
- Privacy and Responsible AI: The system should avoid unnecessary personal identification and use observable pedestrian features rather than facial recognition or identity-based classification. Behavioral predictions must be interpreted as operational labels rather than verified psychological states.

Technical Requirements

The technical requirements define the software tools, data representations, and computational components necessary for implementation.

Table 2:Technical Requirements

Component	**Technical requirement**
Programming language	Python
Dataset	Pedestrian Intention Estimation (PIE) dataset
Pose estimation	MediaPipe
Deep learning	TensorFlow/Keras
Prediction models	LSTM and GRU
Ensemble method	Equal-weight probability averaging
Input features	36 numerical features per frame
Temporal input	30 × 36 feature sequence
Feature preprocessing	Training-fitted feature scaler

Risk assessment	Time-to-Collision calculation
Decision outputs	MAINTAIN, SLOW_DOWN, STOP
Simulation environment	CARLA
Evaluation	Accuracy, precision, recall, F1-score, confusion matrix

The technical design required consistent data exchange between the feature extraction, prediction, and decision-making components. In particular, the inference pipeline needed to preserve the feature structure used during model training to ensure compatibility with the saved LSTM and GRU models.

Design Constraints and Assumptions

The solution was developed within several practical constraints. First, behavioral-state predictions depended on the quality of extracted pedestrian features and available contextual information. Missing or inaccurate pose landmarks could influence classification results.

Second, the six behavioral categories were derived from observable indicators rather than independently verified psychological assessments. Therefore, the framework was intended to model operational behavioral patterns and not directly determine pedestrians' internal mental states.

Third, the decision-making mechanism relied on predefined TTC thresholds and rule-based vehicle responses. These thresholds were used as implementation parameters and were not treated as universally validated safety limits.

Finally, the integrated system was evaluated within CARLA rather than deployed on public roads. Consequently, the solution was designed as a research prototype for investigating socially aware pedestrian prediction and risk-aware decision-making, not as a production-ready autonomous driving safety system.

Overall, these requirements established the functional scope, quality expectations, technical dependencies, and limitations of the proposed solution. They provided the foundation for the system architecture, component design, and integration procedures described in the following sections.

4.2 Component Architecture Within The Complete Solution

The proposed Proactive Social Compliance Modeling Using Theory of Mind component was developed as part of the broader Agentic AI for Robust and Socially Aware Autonomous Driving research project. Its primary purpose was to predict pedestrian behavioral states and support proactive, risk-aware vehicle decision-making during pedestrian–vehicle interactions.

The component architecture consisted of four main processing stages: feature extraction, behavioral-state prediction, risk assessment, and vehicle decision-making. These stages were integrated with the CARLA simulation environment for demonstration and testing.

Table 3:Main Components and Responsibilities

Component	**Responsibility**
Feature Extraction	Extracts 36 pedestrian motion, pose, behavioral, and contextual features.
LSTM–GRU Ensemble	Predicts six pedestrian behavioral states.
Risk Assessment	Evaluates potential conflicts using TTC.
Decision-Making	Generates MAINTAIN, SLOW_DOWN, or STOP actions.
CARLA Integration	Tests the integrated framework in simulated driving scenarios.

Feature Extraction

Pedestrian observations from the PIE dataset were processed to extract 36 numerical features representing motion, body posture, behavioral patterns, and environmental context [1]. MediaPipe was used to obtain pedestrian body landmarks [6]. These features were organized into 30-frame temporal sequences.

Behavioral-State Prediction

Independently trained LSTM and GRU models analyzed the temporal sequences to classify pedestrian behavior into six categories: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk. An equal-weight probability-averaging ensemble combined both models' predictions to produce the final behavioral-state classification and associated confidence value.

Risk Assessment and Decision-Making

The predicted behavioral state was combined with Time-to-Collision (TTC) estimates to assess potential pedestrian–vehicle conflicts. The system used TTC thresholds of 2.5 seconds for warning-level conditions and 1.5 seconds for critical conditions [20]. Based on the estimated risk and predicted behavior, the rule-based decision module generated one of three responses: MAINTAIN, SLOW_DOWN, or STOP.

Integration With the Complete Solution

The component was integrated with the CARLA simulation environment to evaluate pedestrian–vehicle interactions under controlled scenarios [4]. Within the complete autonomous driving solution, this component provided pedestrian behavioral predictions and risk-aware decision support, contributing to the broader objective of socially aware autonomous navigation.

Overall, the modular architecture connected pedestrian perception, temporal behavior prediction, and vehicle decision-making. It enabled the investigation of proactive pedestrian interaction handling while keeping the individual research component separate from the complete autonomous driving system.

4.3 Design Alternatives and Decisions

Several design alternatives were considered during the development of the proposed Theory-of-Mind-inspired pedestrian behavioral-state prediction framework. The final design decisions focused on balancing temporal prediction capability, computational efficiency, interpretability, and compatibility with the CARLA simulation environment.

Feature-Based Representation

An image-based deep learning approach was an alternative for pedestrian behavior prediction. However, the proposed framework adopted a structured feature-based representation consisting of 36 motions, pose, behavioral, and contextual features. This approach was selected to provide interpretable behavioral indicators and reduce dependence on processing complete image sequences during model inference. MediaPipe was used to support pose feature extraction [6].

Temporal Prediction Models

LSTM and GRU networks were selected instead of relying exclusively on static classification models because pedestrian behavior changes over time. Both architectures can learn temporal relationships from sequential observations. Rather than selecting only one model, the framework retained both independently trained networks and combined their class probabilities through equal-weight averaging. This design enabled comparison of individual model performance and ensemble predictions.

Behavioral-State Representation

A binary crossing-intention classification approach was considered less suitable for the research objective because it provides limited information about differences in observable pedestrian behavior. Therefore, six operational behavioral categories—Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk—were used to represent more detailed behavioral patterns. These categories were treated as computational labels rather than verified psychological states.

Risk Assessment and Vehicle Decisions

A rule-based decision mechanism was adopted instead of an end-to-end learned vehicle control policy. The selected approach combined predicted behavioral states with Time-to-Collision (TTC) estimates to generate MAINTAIN, SLOW_DOWN, or STOP decisions. TTC thresholds of 2.5 seconds and 1.5 seconds were used for warning-level and critical conditions, respectively. The 1.5-second critical threshold was informed by traffic-conflict research [20], while the 2.5-second threshold was an implementation choice.

The rule-based approach was preferred because its decision conditions could be examined and interpreted during simulation testing. However, it was not intended to replace a fully validated autonomous vehicle safety controller.

Simulation Environment

CARLA was selected as the integration and evaluation environment because it supports controlled autonomous driving scenarios and pedestrian–vehicle interactions [4]. Simulation-based testing enabled examination of the prediction and decision-making pipeline without introducing experimental risks to real road users.

Overall, the selected design combined structured pedestrian features, LSTM–GRU temporal prediction, probability-based ensemble learning, and interpretable TTC-based decision rules. These decisions supported the research objective of investigating proactive and socially aware pedestrian interaction handling within a simulation-based autonomous driving framework.

4.4 Detailed Design

The proposed framework consists of four main modules: feature extraction, LSTM–GRU behavioral prediction, ensemble classification, and TTC-based vehicle decision-making. These modules were integrated with the CARLA simulation environment [4].

Feature Extraction and Preprocessing

Pedestrian observations from the PIE dataset [1] were processed using MediaPipe pose estimation [6] and feature extraction techniques. A total of 36 features were extracted: 10 motion, 12 pose, nine behavioral, and five contextual features. The features were normalized using a training-fitted scaler and organized into 30-frame sequences with a stride of five, producing an input shape of 30×36.

LSTM–GRU Behavioral Prediction

Two independently trained models, LSTM and GRU, processed the temporal sequences to classify six pedestrian behavioral states: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk. Each model generated a probability distribution across the six classes.

Ensemble Prediction

The final prediction was obtained through equal-weight probability averaging:

$$P_{\text{ensemble}}(c) = \frac{P_{\text{LSTM}}(c) + P_{\text{GRU}}(c)}{2}$$

The class with the highest ensemble probability was selected as the predicted behavioral state, with its probability retained as the confidence value.

Risk Assessment and Vehicle Decision-Making

The predicted behavioral state was combined with Time-to-Collision (TTC) to determine vehicle responses. The system applied the following risk thresholds:

Table 4: TTC thresholds

TTC condition	Risk level	Decision consideration
TTC > 2.5 s	Low	MAINTAIN
1.5 s < TTC ≤ 2.5 s	Warning	SLOW_DOWN
TTC ≤ 1.5 s	Critical	STOP

The predicted behavioral state was also considered when selecting the final action. The 1.5-second critical threshold was informed by traffic-conflict research [20], while the 2.5-second warning threshold was an implementation parameter.

System Integration

The framework was integrated with CARLA to evaluate pedestrian–vehicle interactions under controlled simulation conditions [4]. The processing flow was:

Pedestrian observations → Feature extraction → LSTM and GRU → Ensemble prediction → TTC assessment → Vehicle decision → CARLA simulation.

The six behavioral states represented observable behavioral patterns rather than verified psychological states. The implemented decision rules were evaluated in simulation and were not validated as a real-world autonomous driving safety controller.

4.5 Standards and Professional Practice

The proposed framework followed practical software engineering and machine learning practices to support reliable pedestrian behavioral-state prediction and risk-aware vehicle decision-making.

Modular Software Design: The system separated feature extraction, LSTM–GRU prediction, ensemble inference, and rule-based decision-making into individual modules. This structure supported integration, testing, and maintenance.

Data Processing and Model Evaluation: A track-disjoint 70:15:15 dataset split was used to reduce data leakage. Feature normalization was performed using training data only. Model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrices [17], [18].

Risk-Aware Decision-Making: The system combined predicted pedestrian behavior with Time-to-Collision (TTC) estimates to generate MAINTAIN, SLOW_DOWN, and STOP decisions. The 1.5-second critical-risk threshold was supported by traffic-conflict research [20].

Responsible AI Practice: Pedestrian predictions were based on observable motion, pose, and contextual features rather than personal identification. The six behavioral categories were treated as operational labels, not verified psychological states.

Simulation-Based Testing: CARLA was used to examine the integrated prediction and decision-making framework under controlled pedestrian–vehicle interaction scenarios [4].

These practices supported the development, evaluation, and reproducibility of the proposed research component.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The proposed pedestrian behavioral-state prediction framework was developed using Python as the primary programming language. TensorFlow and Keras were used to implement and train the LSTM and GRU models, while NumPy and scikit-learn supported numerical processing, feature normalization, and model evaluation.

The Pedestrian Intention Estimation (PIE) dataset was used to obtain pedestrian observations for model training and testing [1]. MediaPipe was employed to extract pedestrian body landmarks [6]. The extracted information was processed into 36 motion, pose, behavioral, and contextual features, organized into 30-frame temporal sequences.

The trained LSTM and GRU models were integrated using equal-weight probability averaging to predict pedestrian behavioral states. A Python-based rule-based decision module combined these predictions with Time-to-Collision (TTC) estimates to generate vehicle responses. The complete framework was integrated with the CARLA simulator to examine pedestrian–vehicle interactions under controlled conditions [4].

These technologies supported the implementation, integration, and evaluation of the proposed pedestrian behavioral prediction and risk-aware decision-making framework.

5.2 Component Implementation

The proposed pedestrian behavioral-state prediction component was implemented using Python, integrating feature extraction, temporal deep learning models, ensemble prediction, and risk-aware vehicle decision-making.

Pedestrian observations from the PIE dataset were processed to extract 36 numerical features, including motion, pose, behavioral, and contextual information [1]. MediaPipe was used for body landmark estimation [6]. The extracted features were normalized and organized into 30-frame sequences to capture temporal changes in pedestrian behavior.

Two independent deep learning models, LSTM and GRU, were developed using TensorFlow/Keras to classify six pedestrian behavioral states: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk. Their predicted class probabilities were combined through equal-weight averaging to generate the final ensemble prediction.

The rule-based decision-making module combined predicted behavioral states with Time-to-Collision (TTC) estimates. TTC thresholds of 2.5 seconds and 1.5 seconds were used to identify warning-level and critical conditions, respectively [20]. Based on the predicted behavior and estimated risk, the system generated MAINTAIN, SLOW_DOWN, or STOP decisions.

The implemented components were connected through Python modules and local inference services. The complete prediction and decision-making pipeline was integrated with CARLA to evaluate pedestrian–vehicle interactions under controlled simulation scenarios [4].

Overall, the implementation enabled pedestrian observations to be transformed into behavioral-state predictions and corresponding risk-aware vehicle decisions.

5.3 Integration with The Other Components

The proposed framework was developed and tested independently of the other three research components. Its feature extraction, LSTM–GRU ensemble prediction, TTC-based risk assessment, and vehicle decision-making modules were integrated into a standalone system and demonstrated using CARLA [4].

All four independently developed research components were presented through the "Socialmind Drive" web platform. The platform provided a common interface for showcasing the project components without direct data exchange or technical integration between them.

5.4 Verification and Software Testing

The proposed framework was tested to verify the operation of feature extraction, LSTM–GRU prediction, ensemble classification, and TTC-based decision-making.

Module Testing: Individual Python modules were checked for correct feature processing, input dimensions (30 × 36), model loading, and prediction generation. The preprocessing pipeline was examined to ensure consistent feature normalization.

Model Testing: The LSTM, GRU, and ensemble models were evaluated using a held-out test dataset. Accuracy, precision, recall, F1-score, and confusion matrices were used to assess classification performance [17].

Decision-Making Testing: The rule-based module was examined using TTC warning and critical thresholds of 2.5 and 1.5 seconds, respectively, to assess the generation of MAINTAIN, SLOW_DOWN, and STOP decisions.

Integration Testing: The complete framework was tested in CARLA using controlled pedestrian–vehicle interaction scenarios to observe behavioral predictions, risk estimates, and corresponding vehicle actions [4].

These testing activities supported verification of the implemented components and their integration within the simulation environment.

5.5 Deployment and Operation

The proposed pedestrian behavioral-state prediction framework was deployed as a standalone research prototype within the CARLA simulation environment [4]. The system operated using Python-based modules for feature extraction, LSTM–GRU ensemble prediction, TTC-based risk assessment, and vehicle decision-making.

During operation, pedestrian observations were processed into 30-frame sequences containing 36 features per frame. The trained LSTM and GRU models generated behavioral-state probabilities, which were combined to produce the final prediction. The predicted state and TTC estimate were then used to generate MAINTAIN, SLOW_DOWN, or STOP decisions.

The system was operated and evaluated through controlled CARLA scenarios, with predictions and vehicle responses monitored through the demonstration interface and execution logs. Deployment was limited to simulation-based testing rather than real-world autonomous vehicle operation.

6 RESULTS AND VALIDATION

6.1 Research Results

The proposed pedestrian behavioral-state prediction framework was evaluated using 1,641 held-out test sequences from the Pedestrian Intention Estimation (PIE) dataset. The performance of the LSTM, GRU, and equal-weight LSTM–GRU ensemble models was assessed using classification accuracy, macro-F1-score, and weighted F1-score.

Table 5:Classification Performance of the Proposed Models

Model	Accuracy	Macro-F1	Weighted F1
LSTM	69.8%	59.5%	70.7%
GRU	70.8%	62.6%	71.9%
LSTM–GRU Ensemble	71.5%	62.1%	72.4%

As shown in Table 5, the ensemble achieved the highest classification accuracy of 71.5% and weighted F1-score of 72.4%. This represented accuracy improvements of 1.7 and 0.7 percentage points over the LSTM and GRU models, respectively.

However, the GRU achieved the highest macro-F1-score of 62.6%, indicating slightly better balanced classification performance across the six behavioral categories.

These findings demonstrate that combining LSTM and GRU predictions improved overall classification performance, although the ensemble did not outperform the individual models on every metric.

CARLA Simulation Results

The proposed framework was evaluated in the CARLA simulation environment to examine pedestrian behavioral-state prediction, time-to-collision (TTC)-based risk assessment, and vehicle decision-making [4].

Six single-pedestrian scenarios and one multi-pedestrian scenario were used to demonstrate the system under different pedestrian–vehicle interactions.

Single-Pedestrian Scenario Results

Table 6: Selected Results from Single-Pedestrian CARLA Scenarios

Scenario	Actual State	Predicted State	Confidence	Vehicle Action	Closest Approach
Committed crosswalk	Committed	Committed	97.47%	SLOW_DOWN	4.46 m
Waiting roadside	Waiting	Hesitant	98.41%	STOP	7.14 m
Hesitant pedestrian	Hesitant	Hesitant	96.35%	MAINTAIN	5.94 m
Distracted crossing	Distracted	Committed	99.11%	SLOW_DOWN	3.79 m
Aggressive crossing	Aggressive	Committed	99.64%	SLOW_DOWN	6.60 m
Jaywalking	Jaywalk	Jaywalk	95.00%	SLOW_DOWN	4.13 m

Note: These results represent selected individual scenario executions rather than average performance across repeated trials.

The framework correctly predicted the behavioral states in the selected Committed, Hesitant, and Jaywalk scenarios. However, Waiting was classified as Hesitant, while Distracted and Aggressive were classified as Committed.

The decision-making module generated MAINTAIN, SLOW_DOWN, and STOP responses based on predicted pedestrian behavior and estimated collision risk. These observations demonstrate the functional operation of the framework while identifying limitations in distinguishing certain pedestrian behavioral states.

Multi-Pedestrian Scenario Results

The multi-pedestrian scenario examined the framework's ability to process several pedestrians with different behavioral patterns and select an appropriate overall vehicle response.

Table 7: Observed Multi-Pedestrian Scenario Results

Evaluation Parameter	Observed Result
Pedestrians spawned	16
Maximum simultaneously visible	5
Configured behavioral states	Committed, Waiting, Hesitant, Distracted, Jaywalk
Predicted states observed	Committed, Hesitant
Prediction confidence	45.89%–99.61%
Vehicle actions observed	MAINTAIN, SLOW_DOWN, STOP

The framework processed multiple pedestrian observations and generated individual behavioral-state predictions. The vehicle decision-making module selected the most restrictive required response according to the priority order STOP, SLOW_DOWN, and MAINTAIN.

The results demonstrate the functional capability of the framework to handle multiple pedestrian interactions. However, quantitative classification accuracy and collision-avoidance performance were not established for this scenario.

6.2 Product and Technical Results

The proposed framework was implemented as a standalone pedestrian behavioral-state prediction and risk-aware vehicle decision-making prototype.

The system extracted 36 pedestrian features and organized them into 30-frame sequences for temporal analysis. The trained LSTM and GRU models were integrated into an equal-weight ensemble to predict six operational pedestrian behavioral states: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk.

The TTC-based decision-making module generated three vehicle responses: MAINTAIN, SLOW_DOWN, and STOP. The CARLA implementation demonstrated the connection between behavioral-state prediction, risk assessment, and vehicle decision-making.

The technical implementation produced trained prediction models, a preprocessing scaler, feature extraction modules, inference components, and CARLA scenario execution scripts.

Overall, the implementation demonstrated the technical feasibility of integrating pedestrian behavioral-state prediction with risk-aware vehicle decision-making in a controlled simulation environment.

6.4 Integration Results

The four independently developed research components of Agentic AI for Robust and Socially-Aware Autonomous Driving were presented through the Socialmind Drive web platform.The proposed pedestrian behavioral-state prediction component operated independently, integrating LSTM–GRU ensemble prediction, TTC-based risk assessment, and rule-based vehicle decision-making within CARLA.The web platform provided a common interface for demonstrating the individual research contributions. However, the four components were not technically integrated into a single operational autonomous driving system, and cross-component integration testing was not conducted.

6.5 Summary of Findings

The research demonstrated the feasibility of combining LSTM–GRU ensemble prediction with TTC-based vehicle decision-making.

The ensemble achieved the highest classification accuracy of 71.5% and weighted F1-score of 72.4%, while the GRU achieved the highest macro-F1-score of 62.6%.

The CARLA evaluation demonstrated behavioral-state prediction and vehicle decision-making across six single-pedestrian scenarios and one multi-pedestrian scenario. The framework generated MAINTAIN, SLOW_DOWN, and STOP responses under different pedestrian interactions.

The proposed component was also showcased through the Socialmind Drive platform alongside the other independently developed research components.

Overall, the findings support the technical feasibility of the proposed framework within a simulation environment. However, the behavioral categories represent operational labels rather than verified psychological states, and real-world collision-avoidance or driving safety improvements were not established.

7 DISCUSSION

7.1 Interpretation of Findings

The research findings demonstrate the potential of combining temporal deep learning models with risk-aware decision-making to support socially aware autonomous driving. The LSTM–GRU ensemble achieved 71.5% classification accuracy and 72.4% weighted F1-score, indicating improved overall performance compared with the individual models.

However, the GRU achieved the highest macro-F1-score of 62.6%, compared with 62.1% for the ensemble. This suggests that although ensemble learning improved overall classification performance, it did not consistently improve predictions across all six behavioral categories. Class imbalance and similarities between observable pedestrian behaviors may have contributed to these differences.

The integration of behavioral-state predictions with Time-to-Collision (TTC) assessment enabled the framework to generate MAINTAIN, SLOW_DOWN, and STOP decisions within the CARLA simulation environment [4]. This demonstrated the technical feasibility of connecting pedestrian behavior prediction with interpretable, risk-aware vehicle decision-making.

The findings also highlight an important limitation of the Theory-of-Mind-inspired approach. The six behavioral states were derived from observable pedestrian features rather than independently verified psychological states. Therefore, the results demonstrate behavioral-pattern classification, not direct understanding of pedestrian intentions or mental states.

Overall, the findings support the feasibility of the proposed framework as a simulation-based research prototype. Further improvements in class-level prediction performance and broader validation would be necessary before considering real-world autonomous driving applications.

7.2 Comparison with Literature Baselines and Existing Solutions

Existing pedestrian intention prediction approaches, including PIE [1], PedAST-GCN [9], EfficientPIE [11], and GTransPDM [12], use visual, motion, contextual,

and spatial–temporal information to predict pedestrian crossing behavior. These studies provide important foundations for pedestrian behavior prediction in autonomous driving.

The proposed framework differs by predicting **six operational pedestrian behavioral states** using 36 motion, pose, behavioral, and contextual features. It combines LSTM and GRU models through equal-weight probability averaging and integrates the predicted states with Time-to-Collision (TTC) assessment for risk-aware vehicle decision-making.

Table 8: Comparison with Existing Pedestrian Prediction Approaches

Research approach	**Prediction method**	**Main focus**	**Difference from proposed framework**
PIE [1]	Visual and contextual prediction models	Crossing intention and trajectory prediction	Provides the dataset foundation but uses different prediction targets
PedAST-GCN [9]	Spatial–temporal attention graph convolutional network	Pedestrian crossing intention	Uses graph-based temporal learning rather than an LSTM–GRU ensemble
EfficientPIE [11]	Efficient crossing-intention prediction	Low-latency pedestrian intention estimation	Emphasizes prediction efficiency rather than six-state classification
GTransPDM [12]	Graph-embedded transformer	Pedestrian crossing intention	Uses graph-transformer modeling rather than recurrent ensemble learning

Proposed framework	LSTM–GRU probability-averaging ensemble	Six behavioral states and TTC-based vehicle decisions	Combines multi-class prediction with risk-aware decision support

The proposed ensemble achieved 71.5% accuracy and 72.4% weighted F1-score on the held-out PIE-derived test dataset. The GRU achieved the highest macro-F1-score of 62.6%, indicating that ensemble learning improved overall accuracy but did not outperform the individual models across every metric.

Unlike approaches primarily focused on prediction, the proposed framework also generated MAINTAIN, SLOW_DOWN, and STOP decisions within the CARLA simulation environment [4].

However, direct numerical comparison with published methods is limited because their prediction targets, labeling procedures, and evaluation protocols differ. Consequently, the proposed framework cannot be claimed to outperform these methods based on the reported classification accuracy.

Overall, the main contribution is the integration of multi-class pedestrian behavioral-state prediction, LSTM–GRU ensemble learning, and TTC-based risk-aware decision-making within a standalone autonomous driving research prototype

7.3 Research Contribution

The main contribution of this research is the development of a Theory-of-Mind-inspired pedestrian behavioral prediction framework that combines temporal deep learning with risk-aware autonomous vehicle decision-making.

The research developed a 36-feature behavioral representation covering pedestrian motion, pose, behavioral patterns, and environmental context. These features supported the classification of six operational behavioral states: Waiting, Hesitant,

Committed, Distracted, Aggressive, and Jaywalk, providing a more detailed representation than binary crossing-intention prediction.

An LSTM–GRU ensemble was implemented to combine temporal predictions through equal-weight probability averaging. The ensemble achieved 71.5% accuracy and 72.4% weighted F1-score improving overall classification performance compared with the individual models.

The framework also connected predicted behavioral states with Time-to-Collision (TTC)-based risk assessment generating MAINTAIN, SLOW_DOWN, and STOP vehicle decisions. This extended the research from behavioral prediction to interpretable decision support.

Overall, the research contributes an independently implemented and CARLA-tested framework [4] for pedestrian behavioral-state prediction and risk-aware decision-making. The findings demonstrate technical feasibility within simulation, without claiming direct recognition of psychological states or validated real-world safety improvements.

7.4 Product Contribution

The main product contribution of this research is a standalone pedestrian behavioral-state prediction and risk-aware decision-making prototype developed for autonomous driving simulation.

The implemented system combines MediaPipe-based feature extraction, trained LSTM–GRU models, ensemble prediction, and TTC-based risk assessment to generate MAINTAIN, SLOW_DOWN, and STOP vehicle decisions.

The prototype was integrated with CARLA to demonstrate pedestrian–vehicle interactions under controlled simulation conditions [4]. It also includes reusable trained models, preprocessing tools, inference modules, and scenario execution scripts.

Additionally, the component was presented through the “Socialmind Drive” web platform alongside the other three independently developed research components.

Overall, the product provides a functional research prototype for demonstrating pedestrian behavioral prediction and risk-aware vehicle decision-making within a simulated autonomous driving environment.

7.5 Limitations and Threats to Validity

The proposed framework has several limitations that should be considered when interpreting the research findings.

Dataset and Labeling Limitations: The six pedestrian behavioral states were derived from observable features in the PIE dataset [1] rather than directly annotated psychological states. Therefore, the predicted classes represent operational behavioral categories and cannot confirm pedestrians' actual intentions or mental states.

Model Performance: The ensemble achieved 71.5% accuracy and 62.1% macro-F1-score, indicating that classification errors remain, particularly across less consistently predicted classes. Such errors may affect subsequent vehicle decisions.

Simulation Limitations: The framework was evaluated using CARLA [4]. Simulated pedestrian interactions may not fully represent the complexity, uncertainty, and environmental variations of real-world traffic. Consequently, real-world safety and generalization remain unverified.

Decision-Making Limitations: The TTC-based decision mechanism uses predefined thresholds and rule-based actions. These thresholds were not optimized or validated for all traffic conditions, limiting the framework's adaptability to complex scenarios.

Threats to Validity: The use of a single primary dataset, derived behavioral labels, and simulation-based evaluation limits the generalizability of the findings. Although track-disjoint data splitting was applied to reduce data leakage, further evaluation using independently annotated datasets and real-world scenarios is required.

Overall, these limitations indicate that the proposed framework demonstrates simulation-based technical feasibility, rather than validated real-world safety or direct Theory-of-Mind reasoning.

7.6 Implications

The findings of this research highlight the potential of pedestrian behavioral-state prediction to support socially aware autonomous driving. Predicting six operational behavioral states provides more detailed information about observable pedestrian behavior than binary crossing-intention classification.

The results also demonstrate the practical value of combining LSTM–GRU temporal learning with TTC-based risk assessment. This approach connects pedestrian behavior prediction with interpretable vehicle decisions, supporting more context-aware responses in simulated pedestrian–vehicle interactions.

From a research perspective, the framework provides a foundation for investigating Theory-of-Mind-inspired behavior modeling without assuming direct access to pedestrians' psychological states. It also demonstrates how prediction and decision-making modules can be evaluated together within CARLA [4].

However, the findings are limited to the evaluated dataset and simulation environment. Further validation with diverse real-world data is necessary before the approach can be considered for safety-critical autonomous driving applications.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research developed a Theory-of-Mind-inspired pedestrian behavioral-state prediction and risk-aware decision-making framework to support socially aware autonomous driving.

The proposed framework used 36 motion, pose, behavioral, and contextual features to classify pedestrians into six operational behavioral states. An LSTM–GRU ensemble was implemented to capture temporal behavior patterns, achieving 71.5% accuracy and 72.4% weighted F1-score on the held-out PIE-derived test dataset [1].

The predicted behavioral states were combined with Time-to-Collision (TTC)-based risk assessment to generate MAINTAIN, SLOW_DOWN, and STOP vehicle decisions. The framework was implemented and demonstrated independently within the CARLA simulation environment [4] and presented through the "Socialmind Drive" web platform.

Overall, the research demonstrates the technical feasibility of combining pedestrian behavioral prediction, temporal ensemble learning, and risk-aware decision-making. However, further evaluation using diverse datasets and real-world traffic scenarios is required to establish generalizability and safety effectiveness.

8.2 Achievement of Objectives

The research successfully achieved its main objective of developing a Theory-of-Mind-inspired pedestrian behavioral-state prediction and risk-aware decision-making framework for autonomous driving simulation.

The key research objectives were achieved as follows:

- Pedestrian Behavioral Analysis: Extracted 36 motion, pose, behavioral, and contextual features from the PIE dataset [1] to represent six operational pedestrian behavioral states.

- Behavioral-State Prediction: Developed and evaluated LSTM and GRU models, achieving 71.5% accuracy and 72.4% weighted F1-score through ensemble prediction.
- Risk-Aware Decision-Making: Integrated predicted behavioral states with TTC-based risk assessment to generate MAINTAIN, SLOW_DOWN, and STOP vehicle decisions.
- Simulation and Demonstration: Implemented and demonstrated the standalone framework using CARLA [4] and presented the component through the "Socialmind Drive" web platform.

Overall, the research objectives were achieved at the prototype and simulation level, demonstrating the feasibility of combining pedestrian behavioral prediction with risk-aware autonomous vehicle decision-making. Real-world effectiveness and safety remain areas for further validation.

8.3 Individual Research and Product Contributions

The individual research contribution of this study is the development of a Theory-of-Mind-inspired pedestrian behavioral-state prediction framework for socially aware autonomous driving. The research introduced six operational pedestrian behavioral states using 36 motion, pose, behavioral, and contextual features extracted from the PIE dataset [1].

An LSTM–GRU ensemble was developed to predict pedestrian behavioral states from temporal observations. The ensemble achieved 71.5% accuracy and 72.4% weighted F1-score, demonstrating its effectiveness within the evaluated dataset.

The individual product contribution is a standalone pedestrian behavior prediction and risk-aware decision-making prototype. It integrates feature extraction, trained prediction models, TTC-based risk assessment, and rule-based MAINTAIN, SLOW_DOWN, and STOP decisions within the CARLA simulation environment [4].

The completed component was also presented through the “Socialmind Drive” web platform alongside the other independently developed research components.

Overall, the individual contribution combines multi-class pedestrian behavior prediction with interpretable vehicle decision support, providing a functional simulation-based prototype for further research.

8.4 Future Directions

Future research should focus on improving the accuracy, generalizability, and practical applicability of the proposed pedestrian behavioral-state prediction framework.

Improved Behavioral-State Recognition: Future studies could use larger, more diverse datasets with directly annotated pedestrian behavioral states to improve classification reliability and reduce uncertainty associated with derived labels.

Advanced Prediction Models: Attention-based architecture, transformers, and improved ensemble strategies could be investigated to capture complex pedestrian behavior patterns and improve performance across all six classes.

Adaptive Risk-Aware Decision-Making: The existing TTC-based rule mechanism could be extended with adaptive risk thresholds, prediction confidence, and pedestrian–vehicle interaction modeling to support more context-sensitive decisions.

Real-World Validation: The framework should be evaluated under diverse weather, lighting, traffic, and pedestrian interaction conditions, followed by controlled real-world testing to assess generalization, response reliability, and safety.

Overall, these improvements could support the development of more reliable, adaptive, and socially aware autonomous driving systems while addressing the limitations identified in this research.

REFERENCES

[1] A. Rasouli, I. Kotseruba, T. Kunic, and J. K. Tsotsos, “PIE: A Large-Scale Dataset and Models for Pedestrian Intention Estimation and Trajectory Prediction,” in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 6262–6271.

[2] T. Chen et al., “PSI: A Pedestrian Behavior Dataset for Socially Intelligent Autonomous Car,” arXiv:2112.02604, 2022.

[3] F. Munir and T. P. Kucner, “Context-aware Multi-task Learning for Pedestrian Intent and Trajectory Prediction,” arXiv:2407.17162, 2024.

[4] A. Dosovitskiy et al., “CARLA: An Open Urban Driving Simulator,” in *Proc. Conference on Robot Learning (CoRL)*, 2017, pp. 1–16.

[5] F. Li, S. Fan, P. Chen, and X. Li, “Pedestrian Motion State Estimation From 2D Pose,” manuscript supplied in the research literature, publication details not established.

[6] C. Lugaresi et al., “MediaPipe: A Framework for Building Perception Pipelines,” arXiv:1906.08172, 2019.

[7] B. Varga, T. Brand, M. Schmitz, and E. Hashemi, “Interaction-Aware Model Predictive Decision-Making for Socially-Compliant Autonomous Driving in Mixed Urban Traffic Scenarios,” arXiv:2503.01852v2, 2026.

[8] S. Baron-Cohen, A. M. Leslie, and U. Frith, “Does the autistic child have a ‘theory of mind’?” Cognition, vol. 21, no. 1, pp. 37–46, 1985, doi: 10.1016/0010-0277(85)90022-8.

[9] Y. Ling, Z. Ma, Q. Zhang, B. Xie, and X. Weng, “PedAST-GCN: Fast Pedestrian Crossing Intention Prediction Using Spatial–Temporal Attention Graph Convolution Networks,” IEEE Transactions on Intelligent Transportation Systems, vol. 25, no. 10, Oct. 2024, doi: 10.1109/TITS.2024.3398252.

[10] C. L. Baker, R. Saxe, and J. B. Tenenbaum, “Action Understanding as Inverse Planning,” Cognition, vol. 113, no. 3, pp. 329–349, 2009, doi: 10.1016/j.cognition.2009.07.005.

[11] F. Qu et al., “EfficientPIE: Real-Time Prediction on Pedestrian Crossing Intention with Sole Observation,” in Proc. International Joint Conference on Artificial Intelligence (IJCAI), 2025, pp. 1793–1801.

[12] X. Chen, M. Sun, and J. Liu, “GTransPDM: Graph-Embedded Transformer for Pedestrian Crossing Intention Prediction,” arXiv:2409.20223, 2024.

[13] S. Radhakrishna and A. Balasubramanyam, “MonoPIC—A Monocular Low-Latency Pedestrian Intention Classification Framework for IoT Edges Using ID3 Modelled Decision Trees,” arXiv:2304.00206, 2024.

[14] R. Chandra, A. Bera, and D. Manocha, “StylePredict: Machine Theory of Mind for Human Driver Behavior From Trajectories,” arXiv:2011.04816, 2020.

[15] Z. Zhang, M. F. Elahi, J. Domeyer, and R. Tian, “Driver Temporal Segmentation of Pedestrian Crossing Intentions During Negotiations,” Transportation Research Part F: Traffic Psychology and Behaviour, vol. 114, pp. 953–969, 2025, doi: 10.1016/j.trf.2025.07.002.

[16] F. Munir, S. Azam, T. Mihaylova, V. Kyrki, and T. P. Kucner, “Pedestrian Vision Language Model for Intentions Prediction,” IEEE Open Journal of Intelligent Transportation Systems, vol. 6, pp. 393–406, 2025, doi: 10.1109/OJITS.2025.3554387.

[17] M. Sokolova and G. Lapalme, “A systematic analysis of performance measures for classification tasks,” Information Processing & Management, vol. 45, no. 4, pp. 427–437, 2009, doi: 10.1016/j.ipm.2009.03.002.

[18] H. He and E. A. Garcia, “Learning from imbalanced data,” IEEE Transactions on Knowledge and Data Engineering, vol. 21, no. 9, pp. 1263–1284, 2009, doi: 10.1109/TKDE.2008.239.

[19] J. C. Hayward, “Near-miss determination through use of a scale of danger,” Highway Research Record, no. 384, pp. 24–34, 1972.

[20] R. van der Horst and J. Hogema, “Time-to-Collision and Collision Avoidance Systems,” in Proceedings of the 6th ICTCT Workshop, Salzburg, Austria, 1993.

APPENDICES

Appendix A Academic integrity and AI use disclosure

ChatGPT (OpenAI), Claude (Anthropic), DeepSeek, and Grammarly were used to assist with academic writing, grammar correction, language improvement, content organization, and formatting of this thesis. These tools also supported code debugging and the explanation and presentation of technical information based on the research materials provided.

The research implementation, experimental work, analysis, and findings remain the responsibility of the candidate. AI-assisted content and debugging suggestions were reviewed for accuracy, relevance, and consistency with the research objectives.

The candidate takes full responsibility for the originality, accuracy, and final content of this thesis.